APA-UK007 After Storage and Shipment

What happened?

- The comparable Chicago tension measurements are lower than the UK measurements in every layer-side group.
- The shift is not a single constant offset: G is lower by about `0.2–0.4 N`, while the currently measured U subset is lower by about `0.5–0.9 N`.
- All current Chicago G and U summary values are within the `4.0–8.5 N` tension specification.
- Some apparently large shifts in tension are probably due to measurement errors at Daresbury.

Measurement basis

- UK source data are the APA-UK007 action JSON records: U uploaded `2023-11-21`, G uploaded `2023-12-10`.
  - The exact setup is not known, probably laser+ziptie+labview
- Chicago source data from March 11 2026
  - Chicago winder using laser+compressed air+python
- Residuals are defined as `Chicago - UK`; negative values mean Chicago measured lower tension.

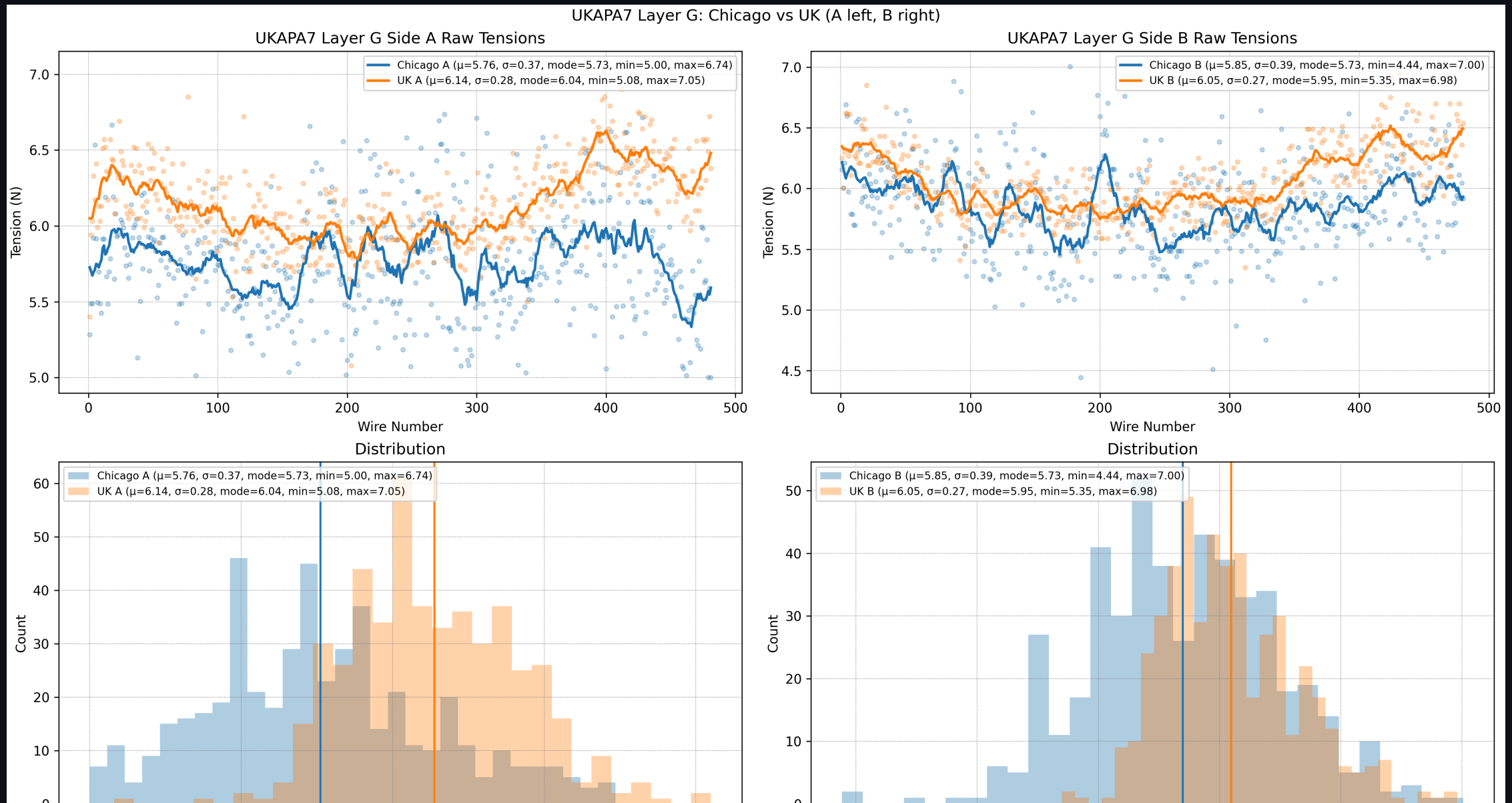
Coverage and limits

- **G layer:** full A-side comparison with `481` wires; corrected B-side comparison with `480` wires after reversing and shifting the B index.
- **U layer:** partial Chicago coverage only, with `385` aligned wires on side A and `641` on side B.
- U layer wires were accessed through a "slit" cut in the G wires near the middle of the APA.
- The G and U comparisons were therefore not made under identical access and frame conditions.

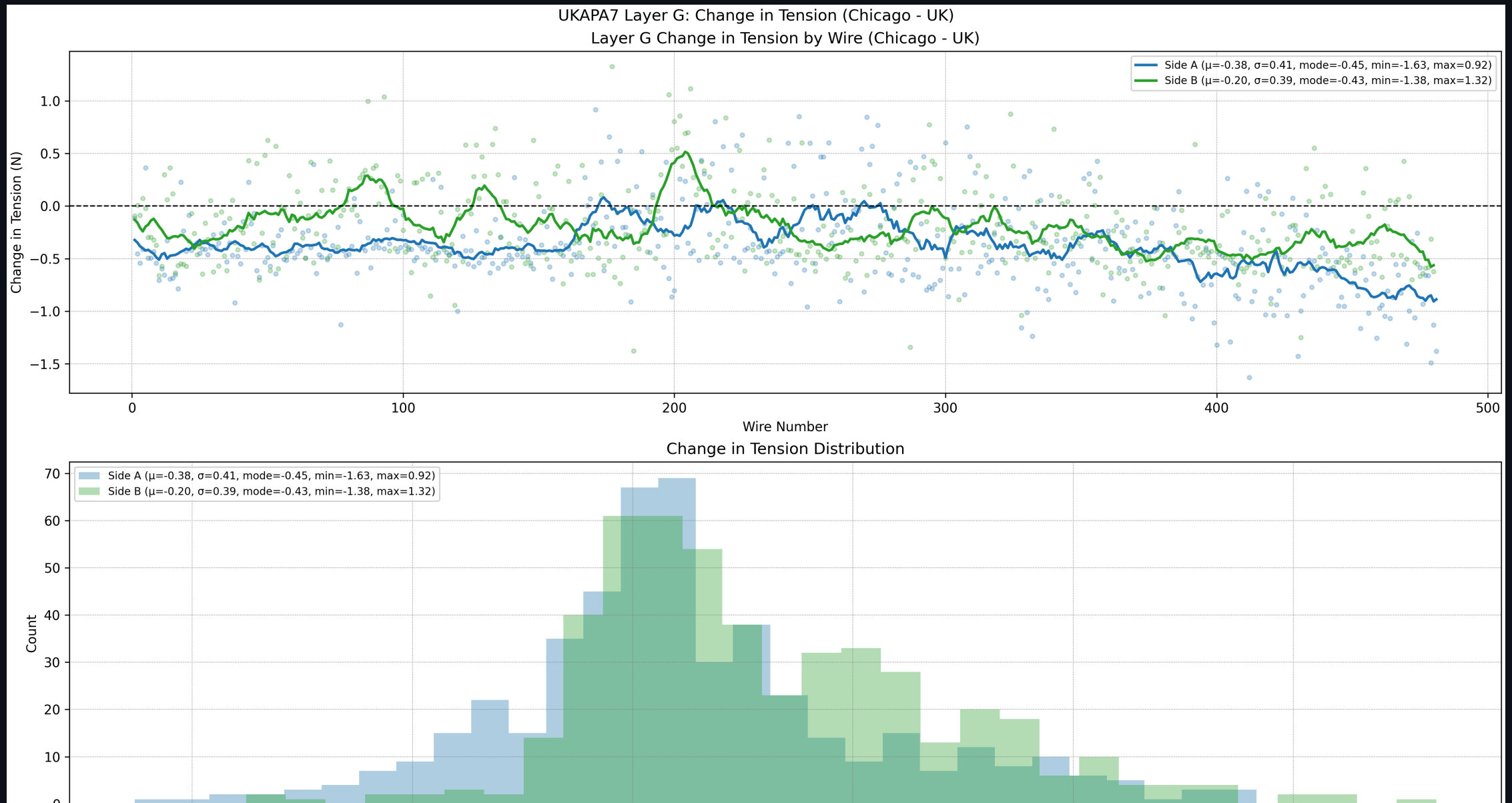
G Layer

- Side A: mean residual `−0.38 N`, mode `−0.45 N`, `84%` of wires lower in Chicago.
- Side B: mean residual `−0.20 N`, mode `−0.43 N`, `71%` of wires lower in Chicago.
- Residual widths are about `0.4 N` on both sides.
- Some individual wires measured higher in Chicago.

Tensions


UKAPA7 Layer G: Chicago vs UK (A left, B right)
UKAPA7 Layer G Side A Raw Tensions
Chicago A (μ=5.76, σ=0.37, mode=5.73, min=5.00, max=6.74)
UK A (μ=6.14, σ=0.28, mode=6.04, min=5.08, max=7.05)
Tension (N)
Wire Number
UKAPA7 Layer G Side B Raw Tensions
Chicago B (μ=5.85, σ=0.39, mode=5.73, min=4.44, max=7.00)
UK B (μ=6.05, σ=0.27, mode=5.95, min=5.35, max=6.98)
Tension (N)
Wire Number
Distribution
Chicago A (μ=5.76, σ=0.37, mode=5.73, min=5.00, max=6.74)
UK A (μ=6.14, σ=0.28, mode=6.04, min=5.08, max=7.05)
Count
Distribution
Chicago B (μ=5.85, σ=0.39, mode=5.73, min=4.44, max=7.00)
UK B (μ=6.05, σ=0.27, mode=5.95, min=5.35, max=6.98)
Count

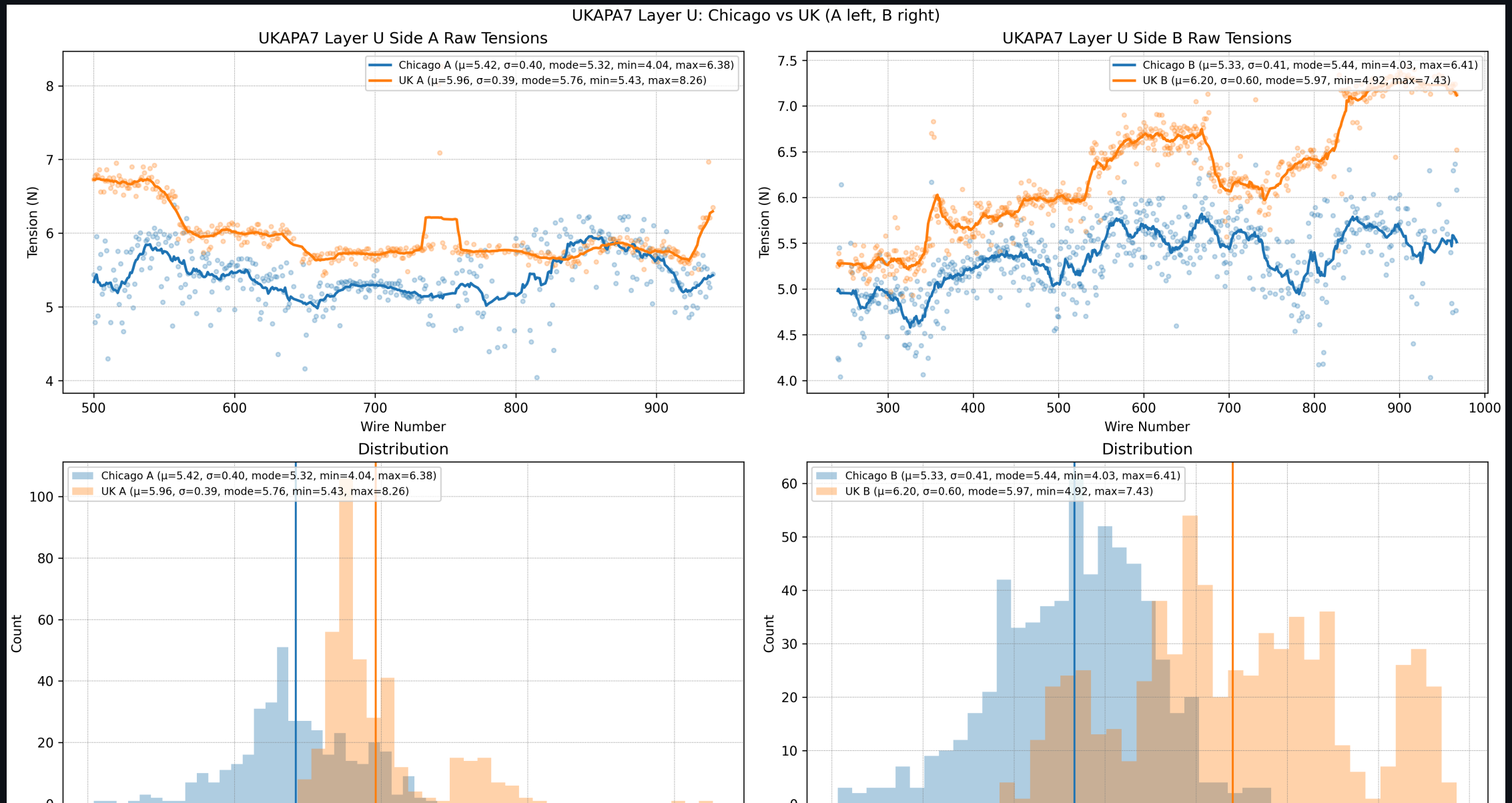
Differences



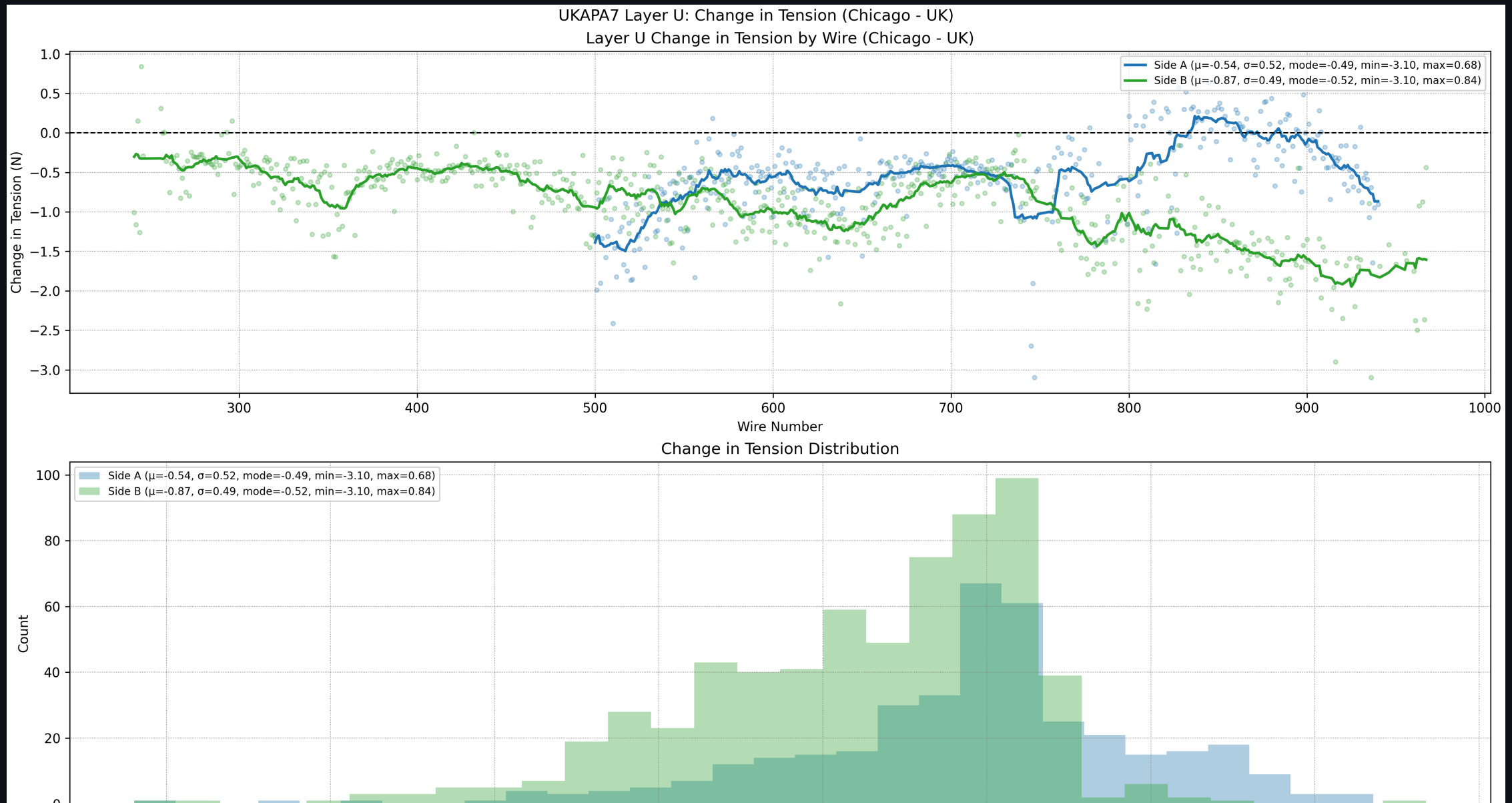
U Layer

- Side A subset: mean residual `-0.54 N`, mode `-0.49 N`, `85%` of aligned wires lower in Chicago.
- Side B subset: mean residual `-0.87 N`, mode `-0.52 N`, `99%` of aligned wires lower in Chicago.
- Residual widths are about `0.5 N` on both sides.
- The U-layer result is a partial-subset comparison, not a full-layer comparison.

Tensions


UKAPA7 Layer U: Chicago vs UK (A left, B right)
UKAPA7 Layer U Side A Raw Tensions
Chicago A (μ=5.42, σ=0.40, mode=5.32, min=4.04, max=6.38)
UK A (μ=5.96, σ=0.39, mode=5.76, min=5.43, max=8.26)
Tension (N)
Wire Number
UKAPA7 Layer U Side B Raw Tensions
Chicago B (μ=5.33, σ=0.41, mode=5.44, min=4.03, max=6.41)
UK B (μ=6.20, σ=0.60, mode=5.97, min=4.92, max=7.43)
Tension (N)
Wire Number
Distribution
Chicago A (μ=5.42, σ=0.40, mode=5.32, min=4.04, max=6.38)
UK A (μ=5.96, σ=0.39, mode=5.76, min=5.43, max=8.26)
Count
Distribution
Chicago B (μ=5.33, σ=0.41, mode=5.44, min=4.03, max=6.41)
UK B (μ=6.20, σ=0.60, mode=5.97, min=4.92, max=7.43)
Count

Differences



Interpretation

- The robust observation is downward shift: Chicago values are lower on average, with a clear peak at `−0.5 N`
- The current Chicago values remain in specification: G ranges from `4.44` to `7.00 N`; U ranges from `4.03` to `6.41 N`.
- The data do not show an out-of-spec tension signature after storage and shipment.

Recommendation

- Tension measurements have long tails, which can erroneously appear as huge shifts in tension.
- As a result, changes alone cannot be the acceptance criterion.
- Establish an acceptance range of tensions after relaxation

Aggregate of all G wires Chicago+Daresbury

all g wires

Aggregate of all U wires Chicago+Daresbury

all u wires